

Fundamental Momentum Based Stock Investing

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Background

Extensive Research on Momentum Strategies

- Explaining the profitability of momentum strategies ([Chan et al., 1996](#))
- Exploring the effectiveness of momentum strategies across diverse asset classes ([Asness et al., 2013](#))
- Constructing momentum factor to describe the cross-sectional variation of commodity returns ([Bakshi et al., 2019](#))

Motivation

Recent Development in Momentum Strategies

- The traditional momentum is based on prices (i.e. **price momentum**).
- **Basis-momentum** factor proposed by [Boons and Prado \(2019\)](#) shows effective performance on predicting commodity premiums in both the time series and cross-section.
- **Macro momentum** strategies developed by [Brooks \(2017\)](#) demonstrates consistent performance over a half-century.
- **Fundamental momentum** strategy proposed by [Novy-Marx \(2015\)](#) provides more stable and significant performance compared with price momentum.

Main Results

Goal of Our Project

We aim to produce new fundamental momentum strategies based on the framework of [Novy-Marx \(2015\)](#), test the effectiveness of the strategy and how they relate to the traditional price momentum strategies, so that we can provide a more comprehensive understanding of momentum strategies.

Empirical Interests

- Strategy performance in back-testing framework.
- Spanning tests.
- Relation with traditional price momentum strategies.
- Conditional strategies.

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Data

Data Sources: WRDS

- **Stock Monthly Data:** CRSP database.
- **Company Quarterly Data:** Compustat database.

Data Process

- **Time Range:** from January 1975 to December 2023.
- Use CUSIP to match the stock price with the fundamental data (24260 stocks in the panel).
- Negative prices are dropped.

Methodology: Backtesting Framework

Quarterly-rebalanced Sorted Portfolio

- At each March, June, September, and December, we compute the fundamental momentum of the selected characteristics.
- If the momentum data is missing when we rebalance the portfolio, we use the most recent data up to two months.
- Considering the Liquidity, we exclude the stocks with the lowest 30% volume.
- Sorting the stocks into deciles based on the fundamental momentum, we construct 10 sorted (equally-weighted) portfolios and hold for 3 months.

Methodology: Profit Margin Momentum

Why Profit Margin?

- Profit margin (PM) is a financial metric that indicates how much of a company's revenue remains as profit after covering all expenses.

$$PM = \frac{\text{Net Income}}{\text{Revenue}} \quad (1)$$

- A higher profit margin indicates a more profitable company that has better control over its costs compared to its competitors.

Methodology: Profit Margin Momentum

Factor Construction

$$\text{PM Momentum}(t) = \frac{PM(t) - PM(t - 3)}{SD(PM_{\text{Last } 3\text{Yrs}})} \quad (2)$$

- To increase the stability, we standardize the profit margin momentum by the standard deviation of the profit margin over the last 3 years.
- Moreover, we require that there are at least 6 observations in the last 3 years to calculate the standard deviation.

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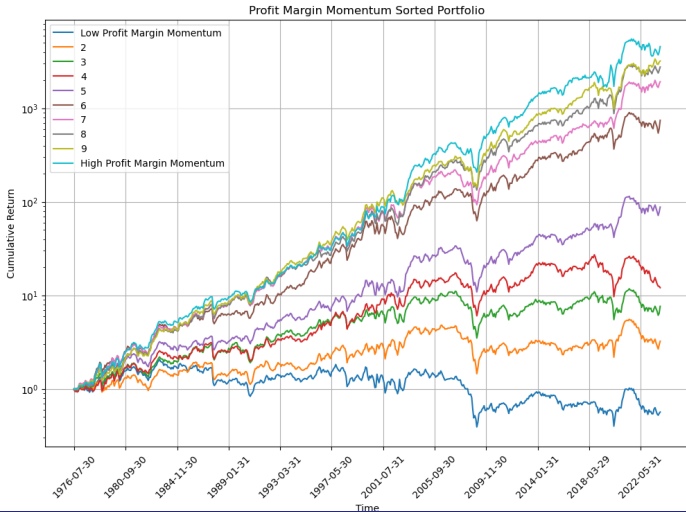
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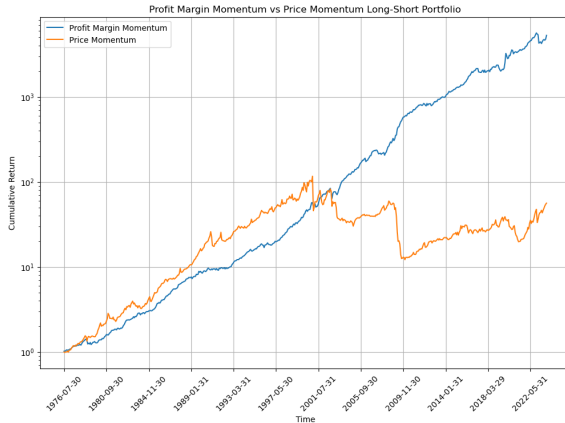
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Sorted Portfolio based on Profit Margin Momentum



Long-short Portfolio based on Profit Margin Momentum



	PM Momentum	Price Momentum
Mean	18.94%	11.76%
Std	12.43%	23.75%
SR	1.52	0.50
MDD	25.01%	89.55%

Spanning Test

Regression Settings

- Regress profit margin momentum on FF3 factors and price momentum.
 - ▶ Examine whether the profit margin momentum can be subsumed.

Regression Results: $R^2=2.60\%$

	α	MKT	SMB	HML	MOM
Coefficient	0.015	-0.000	0.000	0.002	0.021
t-statistic	(9.93)	(-0.73)	(0.88)	(3.59)	(0.96)

Strategy Overlap: Definition

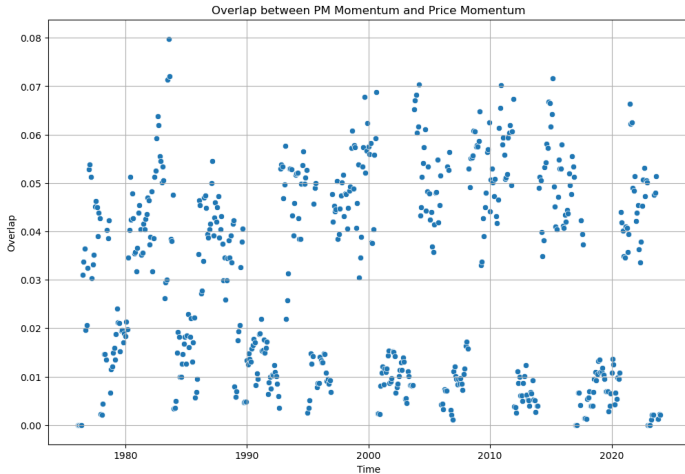
Overlap Ratio

- At time t , for the i -th portfolio of profit margin momentum and price momentum, we calculate the overlap ratio simply by

$$\text{Overlap Ratio} = \frac{\# \text{ of Overlap Stocks}}{\# \text{ of Stocks (Union)}}$$

- For the long-short portfolios, we just take the average of the overlap ratio over the highest and lowest decile portfolios.

Strategy Overlap: Results



Conditional Strategy

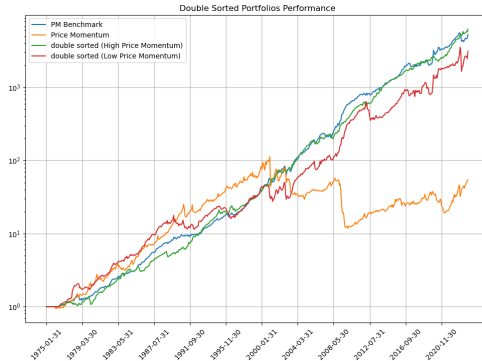
What can be improved?

We want to control the effect of price momentum while building long-short profit margin momentum portfolio. We build long-short double sorted portfolios conditioning on PM momentum and price momentum. We compared following two long-short portfolios:

High PM | High Price – Low PM | High Price

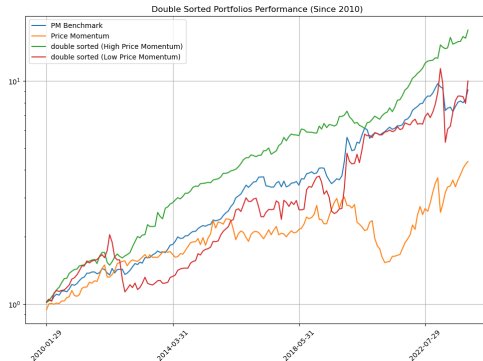
High PM | Low Price – Low PM | Low Price

Conditional Strategy: Results



	PM Momentum	High Price M.	Low Price M.
Mean	18.36%	18.70%	19.00%
Std	12.27%	12.02%	21.90%
SR	1.50	1.56	0.87
MDD	25.01%	23.05%	53.34%

Conditional Strategy: Results(Since 2010)



	PM Momentum	High Price M.	Low Price M.
Mean	16.99%	21.11%	20.96%
Std	14.94%	12.34%	29.34%
SR	1.14	1.71	0.71
MDD	25.01%	16.26%	53.34%

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Conclusion

For double sorted portfolios, controlling the effect of price momentum provdies more stable return.

Reference

Thank you